FitBit Case Study

Data Source

Dataset: Fitbit Fitness Tracker Data

File used: `dailyActivity_merged.csv`

Source: The dataset was made available by Mobius on Kaggle.

License: CC0: Public Domain, as stated in the Coursera case study instructions.

Data period: 12 April 2016 – 12 May 2016.

Description: The dataset contains daily activity records from Fitbit users, including total steps, distance, calories burned, active minutes, and sedentary minutes.

Limitations: The data is historical and comes from a limited group of Fitbit users. Therefore, the findings may not represent all consumers or Bellabeat customers.

Business Task

The purpose of this case study is to analyse smart device usage data from Fitbit users to identify trends in daily activity, including steps taken, active minutes, and sedentary time. As a junior data analyst working with the Bellabeat marketing analyst team, the goal is to understand how consumers use fitness tracking devices and how these trends could apply to Bellabeat customers. The analysis will focus on the Bellabeat Time watch and use the findings to develop high level marketing recommendations that could help Bellabeat better understand its customers and improve its marketing strategy.

Data Cleaning

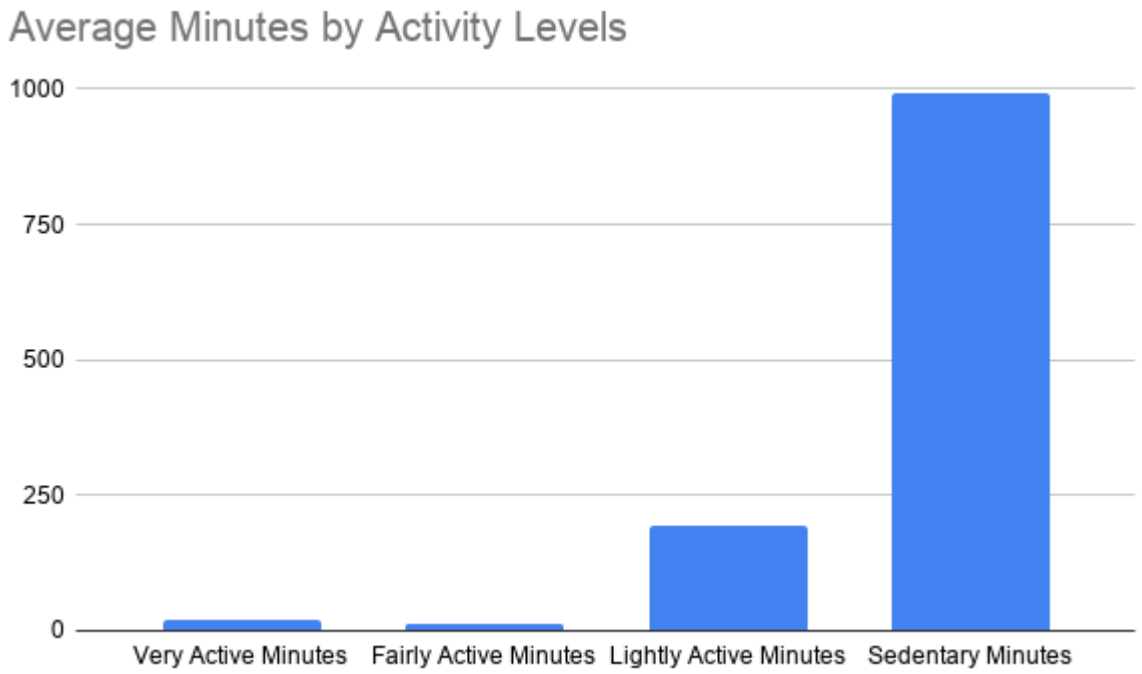
The daily activity dataset was checked for missing values, duplicate user-date records, inconsistent date formats, and negative values. The activity dates were converted into a proper date format for analysis. Records with zero steps were investigated and retained because it was unclear whether they represented inactivity or the device not being worn.

Analysis Process

I used Google Sheets to analyse the cleaned daily activity dataset. I calculated summary statistics, including average, median, minimum, and maximum daily steps, as well as average calories burned and minutes spent at different activity levels. I also created a weekday column and used a pivot table to compare average steps and sedentary minutes across the days of the week. Finally I created charts to visualise the results and identify patterns in user activity.

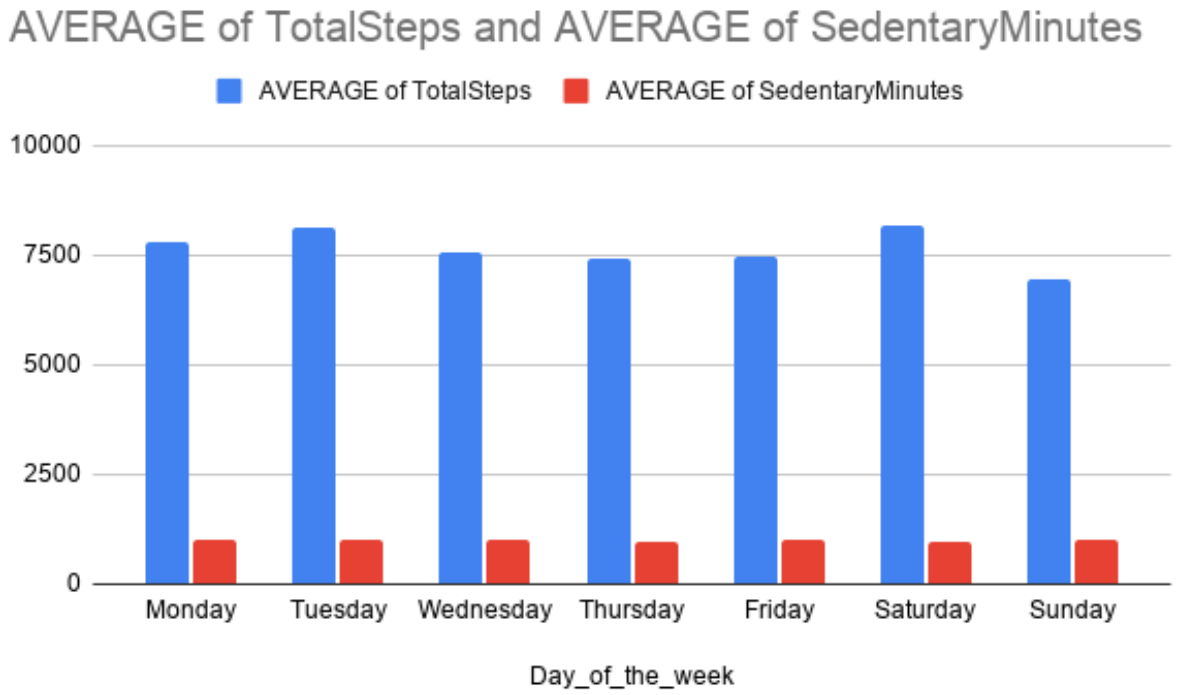
Key Findings

The weekday analysis showed clear differences in recorded activity. Saturday had the highest average daily steps, at approximately 8,153, while Sunday had the lowest, at approximately 6,933. This represents a difference of around 1,220 steps per recorded day. Monday had the highest average sedentary time, at approximately 1,028 minutes, while Thursday had the lowest, at approximately 962 minutes. These findings suggest that activity and sedentary patterns vary throughout the week and may help Bellabeat explore ways to encourage more consistent movement.



Average Minutes by Activity Level

The visualization shows that Fitbit users spend the most time being sedentary, followed by lightly active minutes. Very active and fairly active minutes account for a much smaller portion of their daily activity. This suggests that users may benefit from encouragement to move more throughout the day. Bellabeat could use this insight to promote activity reminders and features that encourage users to build healthier daily movement habits.



Average Daily Steps and Sedentary Minutes by Day of the Week

The visualization compares the average daily steps and average sedentary minutes recorded for each day of the week. Saturday had the highest average daily steps, while Sunday had the lowest. Average sedentary minutes remained relatively consistent throughout the week. These findings suggest that activity levels vary by day, and Bellabeat could use this insight to encourage users to maintain consistent activity and incorporate more movement into their daily routines.

Recommendations

Based on the analysis, I recommend that Bellabeat consider the following marketing strategies:

Encourage Daily Movement: Promote the Bellabeat Time watch's activity-tracking features to encourage users to move more throughout the day, especially on days when average activity is lower.

Promote Movement Breaks: Use marketing messages to encourage users to take regular movement breaks and balance sedentary time with physical activity.

Encourage Consistent Activity: Promote setting personal activity goals and tracking daily progress to help users build consistent, healthy habits.

These recommendations are based on the observed Fitbit usage patterns. Further research would be needed to determine whether the same patterns apply to Bellabeat customers.

Conclusion

The analysis identified differences in daily activity across the week. Saturday had the highest average daily steps at approximately 8,153, while Sunday had the lowest at approximately 6,933. Monday recorded the highest average sedentary time. These findings suggest opportunities for Bellabeat to explore personalised activity goals, movement reminders, and strategies to encourage consistent daily movement. However, the dataset is historical and represents Fitbit users rather than Bellabeat customers, so further research would be needed to confirm whether these trends apply to Bellabeat's target audience.